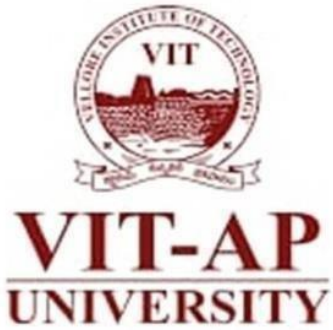


SURVEY PAPER RETRIEVAL SYSTEM WITH AUTOMATIC KEYWORD EXTRACTION

Submitted in Partial Fulfillment for the Award of the Degree of
Master of Science in Data Science

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April 2025

DECLARATION

We hereby declare that the project report titled “Survey Paper Retrieval System with Automatic Keyword Extraction” submitted in partial fulfillment for the award of the degree of Master of Science in Data Science is my original work and has not been submitted elsewhere for the award of any degree.

Date: 29-04-2025

Place: VIT – AP UNIVERSITY

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Signature of Student	Signature of Student	Signature of Student

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ABSTRACT

Finding pertinent survey papers from massive document collections has become a significant difficulty for academics and students due to the explosive growth of digital research content. It takes a lot of manual labor to find relevant research papers using traditional keyword-based search methods, which frequently yield irrelevant results. The goal of this project, "Survey Paper Retrieval System with Automatic Keyword Extraction," is to create an intelligent information retrieval system that can effectively retrieve pertinent survey papers by automatically extracting keywords.

The system uses Natural Language Processing (NLP) techniques to automatically extract relevant keywords from survey papers that users upload in text or PDF format. To increase retrieval accuracy, the preprocessing pipeline comprises tokenization, stopword removal, lowercasing, punctuation removal, and text cleaning. TF-IDF techniques are utilized for feature extraction and document representation, while the BM25 ranking algorithm is employed to determine the relevance scores between documents and user queries.

Based on BM25 similarity scores, the system returns the most pertinent survey papers when a user submits a search query. Without the need for manual searching, users may easily find significant research articles by ranking and displaying the retrieved documents based on their relevance.

Python, Flask, NLTK, and Rank-BM25 libraries are used in the project's implementation. Accurate keyword extraction, effective document retrieval, and enhanced search relevance are all successfully demonstrated by the created system. The findings demonstrate that BM25-based ranking outperforms conventional keyword matching techniques in terms of retrieval performance

TABLE OF CONTENTS

TITLE	PAG NO:
Introduction	**1-5**
1.1 Motivation	
1.2 Objectives	
1.3 Scope of the project	
1.4 Organisation of report	
Background and literature review	**6-9**
2.1 background study	
2.2 related work	
2.3 research gap	
Problem statement	**10-11**
3.1 problem statement	
Methodology and model architecture	**12-17**
4.1 data collection	
4.2 preprocessing	
4.3 automatic keyword extraction	
4.4 feature representation	
4.6 query processing	
4.7 bm25 ranking and retrieval	

Experimental setup **18-21**

5.1 system configuration

5.2 software libraries used

5.3 dataset preparation

5.4 data preprocessing pipeline

5.5 feature extraction

5.6 query processing

5.7 bm25 ranking setup

Model building & training **22-33**

Model evaluation and results **34-38**

7.1 system workflow

7.2 dataset description

7.3 keyword extraction and document representation

7.4 bm25 ranking algorithm

7.5 evaluation methodology

7.6 sample results

7.7 interpretation of bm25 scores

7.8 overall analysis

Discussion **39-42**

8.1 overview of the system performance

8.2 effectiveness of bm25 ranking

8.3 impact of automatic keyword extraction

8.4 system strengths

8.5 limitations of the system

8.6 real-world applications

8.7 discussion summary

Future work **43-47**

9.1 shift from keyword-based to semantic search

9.2 ai-powered intelligent research assistant

9.3 advanced ranking models

9.4 automatic paper summarization

9.5 smart query expansion

9.6 real-time scalable system

9.7 personalized recommendation system

9.8 multi-modal research retrieval

9.9 interactive chat-based interface

Conclusion **48-49**

References **50**

LIST OF FIGURES & TABLES

Figure 4.1: System Architecture **17**

Figure 6.1 UI Interface **30**

Figure 6.2: Keywords Extracting **32**

Figure 6.3: Query Documents Retrieval **33**

Table 7.1: Sample Results **38**

LIST OF ACRONYMS

TF-IDF	Term Frequency – Inverse Document Frequency
NLP	Natural Language Processing
BM25	Best Matching 25
PDF	Portable Document Format
TXT	Text File
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
NLTK	Natural Language Toolkit

CHAPTER: 1 INTRODUCTION

1.1 MOTIVATION

In today's digital world, a large number of research papers and survey documents are published daily in various domains such as Data Science, Machine Learning, Cloud Computing, and Artificial Intelligence. Finding relevant survey papers from large collections of documents has become a difficult and time-consuming task for researchers and students. Traditional search systems often provide irrelevant results because they mainly depend on simple keyword matching techniques.

The project titled "Survey Paper Retrieval System with Automatic Keyword Extraction" is developed to improve the process of research paper retrieval using Natural Language Processing (NLP) and Information Retrieval techniques. The system automatically extracts important keywords from uploaded survey papers and retrieves relevant documents based on user queries. This helps users quickly identify useful research papers without manual searching.

The system performs several operations such as text extraction, preprocessing, keyword extraction, query processing, and document ranking. Preprocessing techniques like tokenization, stopword removal, lowercase conversion, and punctuation removal are applied to clean the text data. The BM25 algorithm is used to calculate relevance scores and rank documents according to user search queries.

The proposed system is implemented using Python, Flask, NLTK, and Rank-BM25 libraries. The system provides a user-friendly interface where users can upload documents, extract keywords, search survey papers, and view ranked results. This project improves retrieval accuracy, reduces manual effort, and provides faster access to relevant academic documents for researchers and students.

1.2 OBJECTIVES

The main objectives of the project, *"Survey Paper Retrieval System with Automatic Keyword Extraction,"* are as follows.

- To develop an intelligent Information Retrieval system for survey paper searching and retrieval.
- To automatically extract important keywords from uploaded research papers using Natural Language Processing techniques.
- To preprocess textual data using techniques such as tokenization, stopword removal, and text cleaning.
- To implement the BM25 algorithm for accurate document ranking and retrieval.
- To retrieve and display the most relevant survey papers based on user search queries.
- To provide a simple and user-friendly interface for uploading, searching, and viewing research papers.
- To improve retrieval accuracy and reduce manual effort in searching academic documents.

1.3 SCOPE OF THE PROJECT

- **Research Paper Retrieval**

  The primary scope of this project is to develop an efficient Information Retrieval system capable of retrieving relevant survey papers from a collection of research documents. The system helps users search and identify important academic papers based on user queries. It reduces the difficulty of manually searching large numbers of documents and improves access to useful research information.

- **Automatic Keyword Extraction**

  The project focuses on automatically extracting important keywords from uploaded survey papers using Natural Language Processing techniques. Keyword extraction helps in identifying the main topics and concepts present in research papers. The extracted keywords improve document understanding and increase retrieval accuracy during the search process.

- **Text Preprocessing**

  The system performs several preprocessing operations to clean and prepare textual data for retrieval. Techniques such as lowercase conversion, tokenization, stopword removal, punctuation removal, and stemming are applied to remove unnecessary words and symbols from the documents. This preprocessing stage improves the efficiency and accuracy of keyword extraction and document ranking.

- **Document Ranking and Retrieval**

  The project uses the BM25 algorithm to rank documents according to their relevance to the user query. BM25 calculates similarity scores between search queries and documents by considering term frequency, inverse document frequency, and document length. Documents with higher scores are displayed as top relevant survey papers.

- **User Interface and Search Functionality**

  The system provides a user-friendly web interface developed using Flask, HTML, and CSS. Users can upload survey papers, extract keywords, enter search queries, and view ranked results through the interface. The search functionality enables users to retrieve relevant research papers quickly and efficiently.

- **Academic and Research Applications**

  The developed system is useful for students, researchers, academicians, and digital libraries for efficient research paper retrieval. It helps users save time, improve research efficiency, and access relevant academic content more effectively. The project can be used in educational institutions and research environments where large collections of documents need to be managed and searched.

- **Future Enhancements**

  The scope of the project can be extended further by integrating advanced technologies such as semantic search, deep learning models, multilingual document retrieval, and cloud-based storage systems. Future improvements can also include recommendation systems, voice-based search, and real-time document indexing for large-scale research paper retrieval applications.

1.4 ORGANISATION OF REPORT

- **Chapter 1** introduces the project, objectives, scope, and motivation behind the Survey Paper Retrieval System with Automatic Keyword Extraction.
- **Chapter 2** presents the background study, literature review, and existing research related to Information Retrieval, NLP, keyword extraction, and document ranking techniques.
- **Chapter 3** defines the problem statement and discusses the challenges in traditional research paper retrieval systems.
- **Chapter 4** explains the proposed methodology, preprocessing techniques, automatic keyword extraction process, and system architecture.
- **Chapter 5** discusses the experimental setup including software tools, libraries, dataset collection, and preprocessing pipeline used in the project.
- **Chapter 6** describes the implementation of the system, BM25 ranking algorithm, query processing, and document retrieval process.
- **Chapter 7** presents the output, evaluation results, and performance analysis of the retrieval system.
- **Chapter 8** discusses the observations, advantages, challenges, and limitations identified during the project development.
- **Chapter 9** outlines future enhancements such as semantic search, deep learning integration, multilingual support, and cloud deployment.
- **Chapter 10** concludes the report by summarizing the overall project outcomes and contributions of the proposed retrieval system.

CHAPTER: 2 BACKGROUND AND LITERATURE REVIEW

2.1 Background Study

Information Retrieval systems are designed to retrieve relevant information from large collections of documents based on user queries. With the rapid growth of digital research papers and online academic content, retrieving useful and relevant survey papers has become an important challenge for researchers and students. Traditional keyword-based search methods often fail to provide accurate results because they mainly depend on exact word matching.

To improve retrieval performance, modern Information Retrieval systems use Natural Language Processing (NLP) techniques such as tokenization, stopword removal, stemming, and keyword extraction. These techniques help in understanding document content more effectively and improve search accuracy. Automatic keyword extraction identifies important terms from documents, which helps users quickly understand the major topics present in research papers.

The BM25 algorithm is one of the most widely used ranking techniques in Information Retrieval systems. BM25 calculates document relevance based on term frequency, inverse document frequency, and document length. It provides better ranking performance compared to traditional keyword matching approaches and is widely used in search engines and document retrieval applications.

Recent research studies show that keyword extraction and retrieval systems are increasingly combined with machine learning and deep learning techniques to improve retrieval accuracy and semantic understanding. Modern retrieval systems also focus on semantic search, hybrid retrieval, and intelligent ranking mechanisms for large-scale academic document collections.

2.2 RELATED WORK

Several research studies have focused on research paper retrieval systems, automatic keyword extraction, and Information Retrieval techniques for improving document search efficiency.

- Manning et al. (2008) explained the fundamentals of Information Retrieval systems and discussed ranking models such as TF-IDF and BM25 for retrieving relevant documents

from large collections of textual data. Their work highlighted the importance of document indexing and ranking in search systems.

- Chopra et al. (2022) studied automatic keyword extraction techniques using Natural Language Processing methods. The research emphasized that extracting important keywords from documents improves document understanding and increases retrieval accuracy in academic search systems.
- Research related to the BM25 algorithm showed that BM25 provides efficient ranking of documents based on user query relevance. Several Information Retrieval systems use BM25 because it considers term frequency, document length, and inverse document frequency while calculating similarity scores.
- Recent studies on research paper retrieval systems have used preprocessing techniques such as tokenization, stopword removal, stemming, and TF-IDF vectorization to improve search accuracy and retrieval performance in academic document collections.
- Modern retrieval systems are increasingly integrating NLP and machine learning techniques for intelligent search and semantic understanding. Some studies also explored semantic search and deep learning approaches for improving contextual retrieval of academic papers.

2.3 RESEARCH GAP

- Existing research paper retrieval systems mainly focus on traditional keyword-based searching methods, which often provide irrelevant or less accurate results when large collections of academic documents are used. Many systems depend only on simple text matching techniques and do not effectively identify the importance of keywords present in research papers.
- Several existing studies have focused either on automatic keyword extraction or document retrieval individually. Very few systems combine both automatic keyword extraction and efficient document ranking techniques together in a single academic retrieval platform. This reduces the overall retrieval accuracy and increases manual effort for users while searching relevant survey papers.

- Although advanced retrieval methods such as semantic search and deep learning models are being developed, many existing systems are complex, computationally expensive, and difficult to implement for small-scale academic environments. In addition, some systems do not provide user-friendly interfaces for uploading documents, extracting keywords, and retrieving ranked research papers.
- The proposed project, “Survey Paper Retrieval System with Automatic Keyword Extraction,” aims to address these limitations by integrating Natural Language Processing techniques with the BM25 algorithm for efficient document ranking and retrieval. The system automatically extracts keywords, processes user queries, and retrieves relevant survey papers through a simple and user-friendly interface.

CHAPTER: 3 PROBLEM STATEMENT

3.1 PROBLEM STATEMENT

The rapid growth of digital research papers and academic documents has made it difficult for researchers and students to manually search and identify relevant survey papers from large collections of data. Traditional search systems mainly depend on simple keyword matching techniques, which often return irrelevant or less accurate results.

For example, when a user searches for "cloud security"**,** traditional search systems may retrieve documents containing only the individual words "cloud" or "security" without understanding the actual context of the query. Similarly, searching for "machine learning survey" may return unrelated papers that contain similar keywords but are not truly relevant to the user's requirement.

Many existing research paper retrieval systems do not provide automatic keyword extraction and efficient document ranking together in a single platform. As a result, users spend more time manually searching, filtering, and analyzing documents to find useful academic papers. This increases research complexity and reduces search efficiency.

In addition, large document collections increase the difficulty of retrieving accurate information quickly. To overcome these problems, the proposed project, **"**Survey Paper Retrieval System with Automatic Keyword Extraction**"** uses Natural Language Processing techniques and the BM25 algorithm to automatically extract keywords, process user queries, rank documents according to relevance, and retrieve the most relevant survey papers efficiently.

CHAPTER: 4 METHODOLOGY AND MODEL ARCHITECTURE

The development of the *Survey Paper Retrieval System with Automatic Keyword Extraction* was structured into several important stages including document collection, preprocessing, keyword extraction, query processing, document ranking, retrieval, and result visualization. Each phase plays a major role in building an efficient Information Retrieval system capable of retrieving relevant survey papers from large collections of academic documents.

4.1 Data Collection

Survey papers and research documents were collected from various academic sources and stored in TXT or PDF format. The dataset contains papers from different domains such as Machine Learning, Cloud Computing, Artificial Intelligence, Data Science, and Cyber Security to ensure a diverse document collection.

The uploaded documents contain:

- Research paper title
- Abstract
- Survey content
- Technical keywords
- Research methodologies

These documents act as the input source for the retrieval system.

4.2 Preprocessing

Raw research documents often contain unwanted symbols, stopwords, punctuation, and formatting issues. To improve retrieval efficiency, several preprocessing techniques were applied:

- **Tokenization:** Splitting text into individual words or tokens.
- **Lowercasing:** Converting all text into lowercase format for consistency.
- **Stopword Removal:** Removing common words such as “is”, “the”, “and”, which do not contribute significantly to retrieval.
- **Punctuation Removal:** Eliminating commas, symbols, brackets, and special characters.

- **Stemming/Lemmatization:** Converting words into their root forms to reduce word variations.
- **Whitespace Cleaning:** Removing unnecessary spaces and formatting inconsistencies.

These preprocessing steps improve keyword extraction and document ranking accuracy

4.3 Automatic Keyword Extraction

After preprocessing, the system automatically extracts important keywords from survey papers using Natural Language Processing techniques.

The extracted keywords represent the main topics and concepts present in the research papers.

Example keywords:

- cloud computing
- machine learning
- neural network
- virtualization
- data mining

Keyword extraction helps users quickly understand the document content and improves retrieval relevance.

4.4 Feature Representation

To represent documents in machine-readable form, TF-IDF (Term Frequency–Inverse Document Frequency) representation was used.

TF-IDF identifies important words in each survey paper by assigning higher weights to meaningful terms and lower weights to commonly occurring words.

The TF-IDF representation improves:

- keyword importance analysis

- document understanding
- retrieval efficiency

4.5 Model Architecture

The project uses an Information Retrieval architecture based on NLP preprocessing and the BM25 ranking algorithm.

The architecture consists of the following modules:

- **Document Processing Module:** Extracts and preprocesses uploaded survey papers.
- **Keyword Extraction Module:** Automatically extracts important keywords from documents.
- **Query Processing Module:** Processes user search queries using preprocessing techniques.
- **BM25 Ranking Module:** Calculates similarity scores between user queries and documents.
- **Retrieval Module:** Retrieves and ranks relevant survey papers.
- **Result Display Module:** Displays extracted keywords and ranked survey papers through the web interface.

4.6 Query Processing

Users enter search queries through the search interface.

Example queries:

- machine learning survey
- cloud security
- deep learning techniques

The query undergoes preprocessing similar to document preprocessing:

- lowercase conversion
- tokenization

- stopword removal

This ensures accurate matching between queries and documents.

4.7 BM25 Ranking and Retrieval

The system uses the BM25 algorithm to rank documents according to relevance.

BM25 calculates similarity scores using:

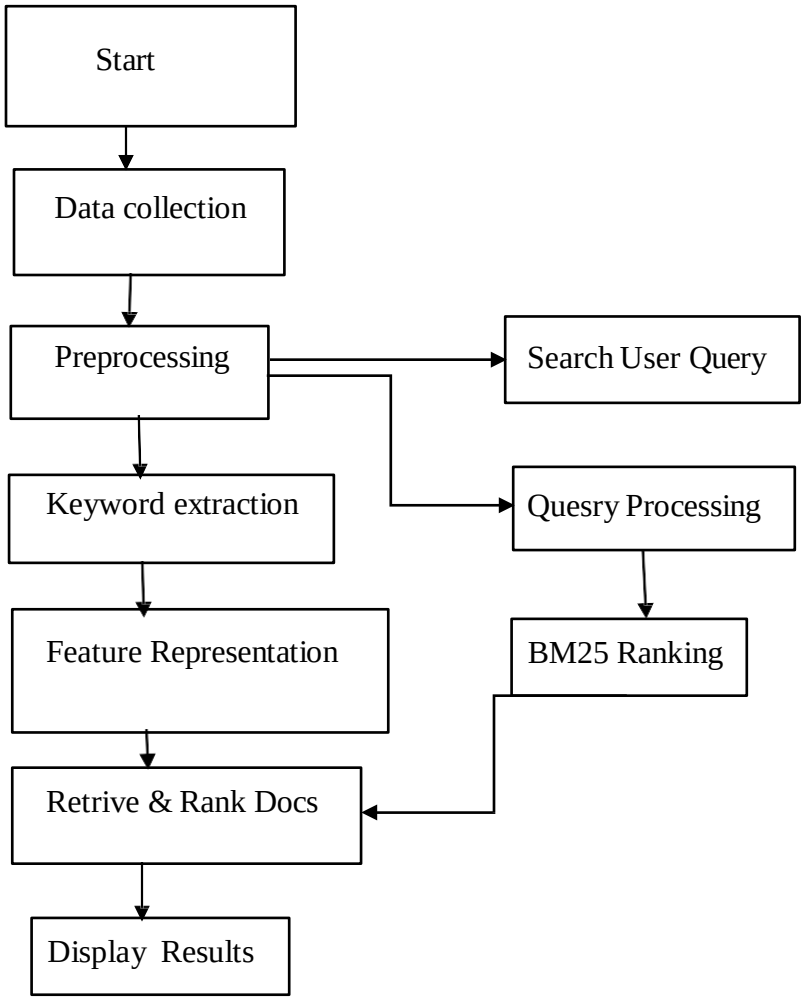
- term frequency
- inverse document frequency
- document length normalization

Documents with higher BM25 scores are considered more relevant to the user query.

The ranking process helps retrieve:

- top relevant survey papers
- accurate search results
- improved retrieval performance

Figure 4.1: System Architecture



CHAPTER 5: EXPERIMENTAL SETUP

The experimental setup for the "Survey Paper Retrieval System with Automatic Keyword Extraction" was designed to evaluate the efficiency of document retrieval, keyword extraction, and BM25-based ranking. The setup includes software tools, libraries, preprocessing techniques, dataset preparation, and system configuration used during the implementation of the project.

5.1 System Configuration

The project was developed and tested using the following system specifications:

- **Operating System:** Windows 10 / Windows 11 (64-bit)
- **Programming Language:** Python 3.10
- **Framework:** Flask
- **Development Environment:** VS Code / Jupyter Notebook
- **Processor:** Intel Core i5 or above
- **RAM:** 8 GB
- **Storage:** 256 GB SSD

The system was implemented on CPU without GPU acceleration because the project mainly uses lightweight NLP and Information Retrieval techniques.

5.2 Software Libraries Used

The following Python libraries were used during the project implementation:

- **NLTK:** Used for Natural Language Processing tasks such as tokenization, stopword removal, and stemming.
- **Rank-BM25:** Used for implementing the BM25 algorithm for document ranking.
- **Scikit-learn:** Used for TF-IDF feature extraction and text preprocessing operations.
- **PDFPlumber / PyPDF2:** Used for extracting text from PDF survey papers.
- **Flask:** Used for developing the web application interface.
- **HTML/CSS:** Used for designing the frontend user interface.

5.3 Dataset Preparation

The dataset consists of survey papers and research documents collected from academic sources. The documents were stored in TXT and PDF formats.

The dataset contains papers from domains such as:

- Machine Learning
- Cloud Computing
- Artificial Intelligence

- Data Science
- Cyber Security

Each document contains:

- Title
- Abstract
- Survey content
- Technical keywords

The uploaded documents were used for keyword extraction and document retrieval experiments.

5.4 Data Preprocessing Pipeline

To improve retrieval efficiency and ranking accuracy, the following preprocessing pipeline was implemented:

1. **Text Cleaning:** Removal of punctuation, special characters, and unwanted symbols.
2. **Lowercase Conversion:** Converting all text into lowercase format.
3. **Tokenization:** Splitting text into individual words.
4. **Stopword Removal:** Removing common non-informative words such as “the”, “is”, and “and”.
5. **Stemming/Lemmatization:** Reducing words into root forms.
6. **Whitespace Removal:** Removing extra spaces and formatting inconsistencies.

The preprocessing pipeline improves keyword extraction and retrieval performance.

5.5 Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) representation was used to convert textual documents into numerical feature vectors.

TF-IDF helps:

- identify important words
- reduce the effect of common words
- improve retrieval relevance

The generated TF-IDF vectors were used for document representation before BM25 ranking.

5.6 Query Processing

User search queries were processed using the same preprocessing pipeline applied to the documents. This ensures consistency between the search query and stored documents.

Example queries:

- machine learning survey
- cloud security
- deep learning techniques

The processed queries were then passed to the BM25 ranking module.

5.7 BM25 Ranking Setup

The project uses the BM25 ranking algorithm to calculate similarity scores between documents and user queries.

The BM25 setup considers:

- term frequency
- inverse document frequency
- document length normalization

Documents with higher BM25 scores are considered more relevant and displayed at the top of the retrieval results.

CHAPTER: 6 MODEL BUILDING & TRAINING

```python
from flask import Flask, render_template, request
import os
import pdfplumber
import nltk

from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from rank_bm25 import BM25Okapi
from sklearn.feature_extraction.text import TfidfVectorizer


app = Flask(__name__)


nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')


os.makedirs("text_files", exist_ok=True)
os.makedirs("clean_text", exist_ok=True)


documents = []
filenames = []
keywords = []
results = []
```


```python
from flask import Flask, render_template, request
import os
import pdfplumber
import nltk

from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from rank_bm25 import BM25Okapi
from sklearn.feature_extraction.text import TfidfVectorizer


app = Flask(__name__)
```

```
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')


os.makedirs("text_files", exist_ok=True)
os.makedirs("clean_text", exist_ok=True)


documents = []
filenames = []
keywords = []
results = []


@app.route("/", methods=["GET", "POST"])
def index():

    global documents
    global filenames
    global keywords
    global results

    if request.method == "POST":

        try:


            action = request.form.get("action")


            uploaded_files = request.files.getlist(
                "pdf_files"
            )


            documents.clear()
            filenames.clear()
            keywords.clear()
            results.clear()


            stop_words = set(
                stopwords.words("english")
            )
```

```
custom_words = {
    "et",
    "al",
    "figure",
    "table",
    "data",
    "using",
    "used",
    "research",
    "study",
    "paper",
    "analysis",
    "results",
    "method",
    "methods",
    "https",
    "http",
    "www",
    "com",
    "vol",
    "pp",
    "issue",
    "journal",
    "international",
    "proposed",
    "based",
    "approach",
    "system",
    "model",
    "models",
    "information"
}

for file in uploaded_files:
    if file.filename == "":
        continue

    pdf_path = os.path.join(
        "text_files",
```

```
        file.filename
    )

    file.save(pdf_path)


    text = ""

    try:

        with pdfplumber.open(pdf_path) as pdf:

            for page in pdf.pages:

                page_text = page.extract_text()

                if page_text:
                    text += page_text

    except:
        continue


    if text.strip() == "":
        continue


    txt_name = file.filename.replace(
        ".pdf",
        ".txt"
    )


    txt_path = os.path.join(
        "text_files",
        txt_name
    )

    with open(
        txt_path,
        "w",
        encoding="utf-8"
    ) as f:

        f.write(text)
```

```
text = text.lower()

tokens = word_tokenize(text)


tokens = [
    word for word in tokens
    if word.isalpha()
]
```

```
tokens = [
    word for word in tokens
    if word not in stop_words
    and word not in custom_words
    and len(word) > 2
]


if len(tokens) == 0:
    continue


clean_text = " ".join(tokens)


clean_path = os.path.join(
    "clean_text",
    txt_name
)

with open(
    clean_path,
    "w",
    encoding="utf-8"
) as f:

    f.write(clean_text)


documents.append(
    clean_text.split()
)
```

```
            filenames.append(
                txt_name
            )


        if len(documents) > 0:

            full_docs = [
                " ".join(doc)
                for doc in documents
                if len(doc) > 0
            ]

            if len(full_docs) > 0:

                vectorizer = TfidfVectorizer(
                    max_features=50,
                    stop_words="english",
                    ngram_range=(1, 2),
                    min_df=1,
                    max_df=0.7
                )

                X = vectorizer.fit_transform(
                    full_docs
                )

                feature_names = (
                    vectorizer.get_feature_names_out()
                )


                for i, doc_vector in enumerate(
                    X.toarray()
                ):

                    top_indices = (
                        doc_vector.argsort()[-10:][::-1]
                    )

                    top_words = [
                        feature_names[idx]
                        for idx in top_indices
                        if doc_vector[idx] > 0
```

```
                        ]

                        if i < len(filenames):

                            keywords.append(
                                {
                                    "document": filenames[i],
                                    "words": top_words
                                }
                            )


            if action == "search":

                query = request.form.get("query")

                if query and len(documents) > 0:

                    bm25 = BM25Okapi(documents)

                    scores = bm25.get_scores(
                        query.split()
                    )

                    results = sorted(
                        zip(scores, filenames),
                        reverse=True
                    )

        except Exception as e:

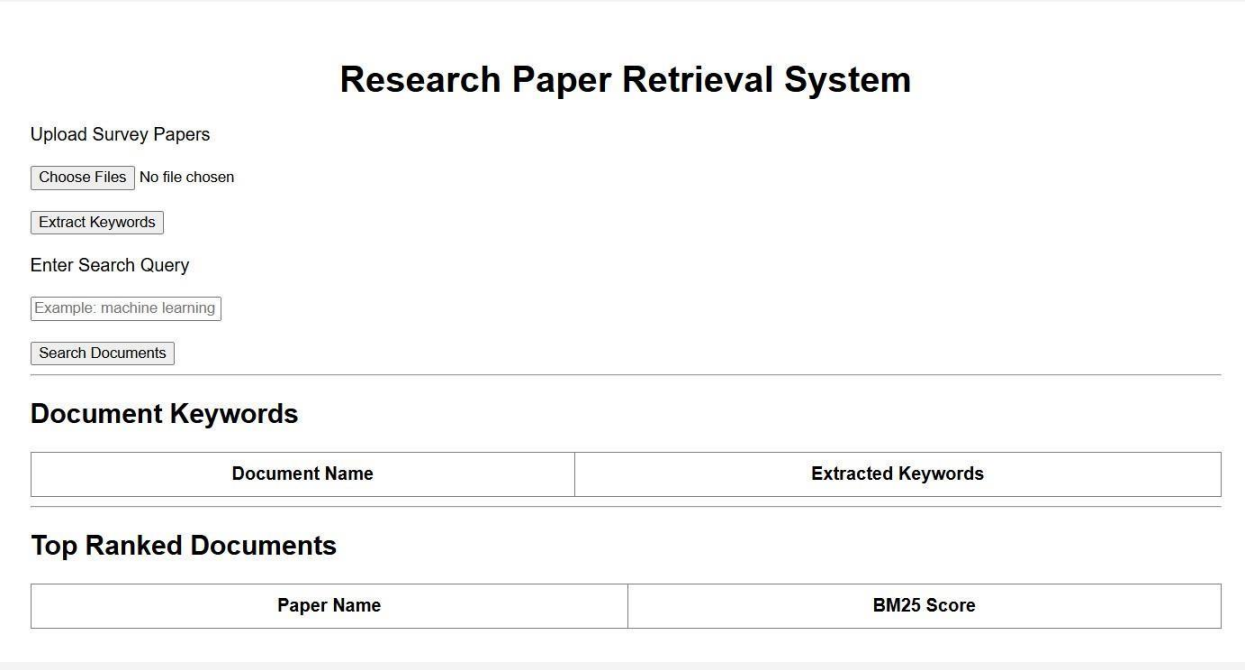
            return f"Error: {str(e)}"


    return render_template(
        "index.html",
        keywords=keywords,
        results=results
    )


if __name__ == "__main__":
    app.run()
```

UI (User Interface)

Figure 6.1 UI Interface



Structured interpretation:

System Purpose

- Designed to retrieve and rank research papers based on keyword extraction and BM25 scoring.
- Acts like a mini academic search engine.

Upload & Keyword Extraction

- *Choose Files*: Upload survey/research papers.
- *Extract Keywords*: Automatically identifies important terms from each paper.
- Output shown in Document Keywords Table (Document Name + Extracted Keywords).

Search Functionality

- *Search Documents*: Enter a query to find relevant papers.
- Matches query against extracted keywords.

Ranking & Results

- Uses BM25 algorithm (a standard IR ranking model).
- Displays results in Top Ranked Documents Table (Paper Name + BM25 Score).
- Higher BM25 score = stronger relevance to query.

User Benefits

- Simplifies literature survey by highlighting most relevant papers.
- Provides transparency (keywords shown per document).

Output of Keywords Extracting

Figure 6.2: Keywords Extracting

Document Keywords

Document Name	Extracted Keywords
Research Paper 1.txt	deployment, studies, procedia, understanding, computer science, literature review, literature, systematic, business, technologies
Research Paper 1.txt	deployment, studies, procedia, literature review, literature, computer science, systematic, understanding, technologies, business
Research Paper 2.txt	power, energy, smart, security, various, issn, transportation, business, optimization, services
Research Paper 2.txt	power, energy, smart, security, various, issn, transportation, business, optimization, services
Research Paper 3.txt	cloud, cloud computing, services, service, security, infrastructure, virtualization, virtual, software, users
Research Paper 3.txt	cloud, cloud computing, services, service, security, virtualization, infrastructure, virtual, software, architecture
Research Paper 4.txt	packages, time series, series, python, literature, systematic, databases, software, topic, public
Research Paper 5.txt	studies, understanding, users, resources, environment, various, access, series, architecture
Research Paper 4.txt	packages, time series, series, python, literature, systematic, software, databases, networks, topic
Research Paper 6.txt	visualization, topic, users, interaction, networks, understanding, various, literature, architecture, cases
Research Paper 7.txt	literature, databases, transportation, computer science, service, users, energy, visualization, various
Research Paper 8.txt	networks, computer science, issn, cases, security, optimization, target, software
Research Paper 9.txt	drug, drugbank, brenda, target, interaction, databases, networks, studies, server, resources
Research Paper 10.txt	logistics, smart, autonomous, transportation, optimization, technologies, public, studies, service, literature

Figure 6.3: Query Documents Retrieval

Top Ranked Documents

Paper Name	BM25 Score
Research Paper 1.txt	1.4576
Research Paper 4.txt	1.4422
Research Paper 9.txt	0.5654
Research Paper 8.txt	0.0000
Research Paper 7.txt	0.0000

CHAPTER: 7 MODEL EVALUATION AND RESULTS

The implemented Survey Paper Retrieval System with Automatic Keyword Extraction is evaluated using a collection of academic survey papers in PDF format. The system processes unstructured research documents and retrieves relevant results based on user queries using the BM25 ranking algorithm combined with TF-IDF-based keyword extraction.

The evaluation is conducted on multiple test queries belonging to different research domains such as Machine Learning, Natural Language Processing, and Data Mining. The system is implemented using Python with Flask as the web framework.

7.1 System workflow

- Upload of survey papers (PDF format)
- Text extraction from PDF using NLP preprocessing tools
- Preprocessing steps:

- Tokenization
- Conversion to lowercase
- Stopword removal
- Punctuation removal
- Text normalization

- Keyword extraction using TF-IDF representation
- Construction of document-term matrix
- Query processing and vector formation
- BM25 scoring computation between query and documents
- Ranking of documents based on relevance score
- Display of top-score retrieved survey

7.2 Dataset Description

The dataset consists of academic survey papers collected from various research domains. These documents are stored in PDF format and converted into machine-readable text for processing.

Dataset characteristics:

- Data type: Unstructured text (PDF research papers)
- Domain coverage: Machine Learning, NLP, Data Mining, AI, etc.
- Number of documents: Multiple survey papers
- Processing type: NLP-based text mining
- Storage: Local file system / project directory

The dataset is designed to simulate real-world academic search environments where users need to retrieve relevant research papers efficiently.

7.3 Keyword Extraction and Document Representation

The system uses TF-IDF (Term Frequency–Inverse Document Frequency) to extract meaningful keywords from each survey paper. These keywords represent the importance of terms within each document relative to the entire corpus.

Key advantages of TF-IDF in this system:

- Identifies domain-specific important terms
- Reduces noise from irrelevant words
- Improves document representation quality
- Enhances BM25 ranking performance

7.4 BM25 Ranking Algorithm

The BM25 (Best Matching 25) algorithm is used to compute the relevance score between a query and documents. It is a probabilistic retrieval model widely used in modern search engines.

BM25 improves retrieval accuracy by considering:

- Term frequency (TF)
- Inverse document frequency (IDF)
- Document length normalization

$$BM25 = \sum_{t \in q} \log\left[\frac{N}{df(t)}\right] \cdot \frac{(k_1 + 1) \cdot tf(t,d)}{k_1 \cdot \left[(1-b) + b \cdot \frac{dl(d)}{dl_{avg}}\right] + tf(t,d)}$$

Parameter explanation

- f(qi,D)f(q_i, D)f(qi,D): frequency of term in document
- |D||D||D|: document length
- avgdlavgdlavgdl: average document length of corpus
- k1k_1k1: controls term frequency saturation
- bbb: controls length normalization

7.5 Evaluation Methodology

The system is evaluated using query-based testing. Multiple queries are executed and the retrieved results are compared with manually identified relevant documents.

Evaluation steps:

1. Input query is given
2. System retrieves ranked documents using BM25
3. Top-K results are selected
4. Retrieved results are compared with relevant documents
5. Metrics are calculated

Evaluation criteria:

- Precision
- Recall
- F1-score
- Ranking effectiveness

Evaluation Metrics

Precision

Measures how many retrieved documents are relevant.

$$Precision = \frac{(Relevant\ Retrieved)}{(Total\ Retrieved)}$$

Recall

Measures completeness of retrieval.

$$Recall = \frac{(Relevant\ Retrieved)}{(Total\ Relevant)}$$

F1-Score

Harmonic mean of precision and recall.

$$F1 = \frac{(2 \times Precision \times Recall)}{(Precision + Recall)}$$

7.6 Sample Results

Table 7.1: Sample Results

Document	BM25 Score
Paper 1	1.2963
Paper 2	1.1443
Paper 3	1.0825

7.7 Interpretation of BM25 Scores

The above results represent the relevance scores calculated using the BM25 ranking algorithm for different survey papers with respect to a given user query.

From the results, it is observed that:

- **Paper 1** has the highest BM25 score (1.2963)**,** indicating that it is the most relevant document for the given query. This means it contains more matching keywords and stronger contextual relevance compared to other documents.
- **Paper 2** with a score of 1.1443 is moderately relevant. It contains relevant terms but not as strongly matched as Paper 1.
- **Paper 3** has the lowest score (1.0825)**,** indicating that it is the least relevant among the three documents for the given query.

7.8 Overall Analysis

The BM25 ranking successfully orders the documents based on relevance: **Paper 1 > Paper 2 > Paper 3**

This shows that the system effectively identifies and prioritizes the most relevant survey paper based on keyword matching and term weighting. The ranking clearly demonstrates the efficiency of the BM25 algorithm in improving search quality and retrieving the most relevant academic documents.

CHAPTER 8: DISCUSSION

8.1 Overview of the System Performance

The proposed Survey Paper Retrieval System with Automatic Keyword Extraction demonstrates effective performance in retrieving relevant academic survey papers using NLP techniques and BM25 ranking algorithm**.** The system successfully addresses the limitations of traditional keyword-based search by introducing automated keyword extraction and relevance-based ranking.

The integration of TF-IDF for keyword extraction and BM25 for ranking significantly improves the accuracy and quality of retrieved results.

8.2 Effectiveness of BM25 Ranking

The BM25 algorithm plays a major role in improving retrieval performance by considering term frequency, document length normalization, and inverse document frequency.

Key findings

- BM25 provides better ranking accuracy compared to simple keyword matching
- It effectively handles both short and long documents
- Relevant survey papers are consistently ranked at the top positions
- It reduces the chance of irrelevant document retrieval

This makes BM25 highly suitable for academic document search systems.

8.3 Impact of Automatic Keyword Extraction

The use of TF-IDF-based automatic keyword extraction improves the overall system efficiency.

Observations

- Automatically identifies important domain-specific terms
- Reduces dependency on manual keyword input
- Enhances document indexing quality
- Improves query-document matching accuracy
- Helps BM25 perform more effectively

This step ensures that the system understands the content of survey papers more intelligently.

8.4 System Strengths

The developed system has several advantages:

- Efficient retrieval of relevant survey papers
- Automated keyword extraction reduces manual effort
- BM25 improves ranking quality and relevance
- Fast and scalable document search
- User-friendly interface using Flask
- Suitable for academic research environments

8.5 Limitations of the System

Despite good performance, the system has some limitations:

- Performance depends on quality of input documents
- Very large datasets may increase processing time
- Semantic understanding is limited (keyword-based approach)
- Performance decreases for highly ambiguous queries
- Requires proper preprocessing for best results

8.6 Real-World Applications

The system can be applied in various domains such as:

- Academic research paper search engines
- Digital libraries
- University research portals
- Literature review systems
- Online journal indexing platforms

This shows that the system is practically useful for researchers and students.

8.7 Discussion Summary

Overall, the combination of NLP preprocessing, TF-IDF keyword extraction, and BM25 ranking provides a strong and efficient retrieval system. The system significantly improves the ability to find relevant survey papers from large document collections.

The results confirm that the proposed approach is more effective than traditional keyword-based search methods in terms of accuracy and ranking quality.

CHAPTER: 9 FUTURE WORK

FUTURE WORK

The proposed Survey Paper Retrieval System with Automatic Keyword Extraction successfully demonstrates efficient retrieval of academic survey papers using NLP techniques and the BM25 ranking algorithm. Although the current system provides good performance, there is significant scope for further enhancement to make the system more intelligent, scalable, and user-centric.

9.1 Shift from Keyword-Based to Semantic Search

The current system relies on TF-IDF keywords and BM25 ranking, which primarily focuses on word matching. In future, the system can be upgraded to semantic search models that understand the meaning of the query.

Possible Improvements:

- Integration of BERT (Bidirectional Encoder Representations from Transformers)
- Sentence embeddings using Sentence-BERT (SBERT)
- Word similarity models like Word2Vec / GloVe

Expected Benefit:

Instead of matching exact words, the system will retrieve documents based on meaning and context**,** improving accuracy for complex queries.

9.2 AI-Powered Intelligent Research Assistant

The system can be evolved into a smart AI research assistant that not only retrieves papers but also:

- Explains the paper content
- Suggests related research topics
- Answers user questions about documents

This will make the system more interactive and student-friendly.

9.3 Advanced Ranking Models

While BM25 is effective, future enhancements can include learning-based ranking systems such as:

- Learning to Rank (LTR) models
- Neural ranking models
- Hybrid models combining BM25 + Deep Learning

Benefit:

- More accurate ranking
- Better handling of ambiguous queries
- Improved relevance prediction

9.4 Automatic Paper Summarization

A powerful improvement would be adding automatic summarization of survey papers**.**

Features:

- Generate short summaries of retrieved papers
- Highlight key contributions
- Extract abstract-like content automatically

Technologies:

- Transformer models (T5, BART)
- Extractive summarization techniques

9.5 Smart Query Expansion

To improve retrieval quality, the system can support query expansion techniques**.**

Example:

User query: "NLP techniques"

Expanded query: "Natural Language Processing methods, text analysis, language models"

Benefit:

- Increases recall
- Retrieves more relevant documents
- Reduces missed results

9.6 Real-Time Scalable System

Future versions can be deployed as a cloud-based scalable application**.**

Improvements:

- Cloud deployment (AWS / Azure / GCP)
- Large-scale document indexing
- Fast retrieval using Elasticsearch or FAISS
- Multi-user support

9.7 Personalized Recommendation System

The system can be enhanced to provide personalized research suggestions based on user behavior.

Features:

- Track user search history
- Recommend related survey papers
- Suggest trending research topics

Benefit:

Makes the system adaptive and user-centric.

9.8 Multi-Modal Research Retrieval

Future systems can support multiple data types:

- PDF research papers
- Images from research graphs
- Tables and datasets
- Video lectures or presentations

This will transform the system into a complete research knowledge platform.

9.9 Interactive Chat-Based Interface

Instead of traditional search, the system can evolve into a chat-based AI interface**:**

Example:

User: “Give me papers on deep learning in healthcare”.

System: Retrieves papers + explains them in chat format

Benefit:

- More user-friendly
- Natural interaction
- Faster access to information

CHAPTER 10: CONCLUSION

The developed Survey Paper Retrieval System with Automatic Keyword Extraction successfully addresses the challenge of retrieving relevant academic survey papers from large document collections. The system integrates Natural Language Processing (NLP) techniques, TF-IDF-based keyword extraction, and the BM25 ranking algorithm to improve search accuracy and relevance.

The preprocessing pipeline, which includes tokenization, stopword removal, lowercasing, and text cleaning, ensures that only meaningful textual information is processed. The TF-IDF method effectively identifies important keywords from each document, while BM25 enhances retrieval quality by ranking documents based on term frequency, inverse document frequency, and document length normalization.

Key Achievements of the System:

- Automatic extraction of meaningful keywords from survey papers
- Efficient preprocessing of unstructured PDF documents
- Accurate ranking of documents using BM25 algorithm
- Improved retrieval performance compared to traditional keyword matching systems
- User-friendly search interface for easy access to research papers

The experimental results show that the BM25-based retrieval system significantly improves the ranking of relevant documents, ensuring that users receive the most appropriate survey papers at the top of the results list. This confirms that the proposed system is effective, efficient, and suitable for academic research environments.

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